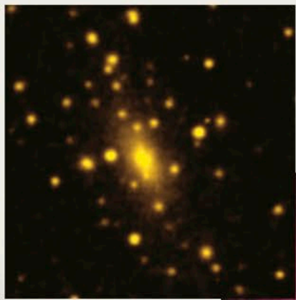
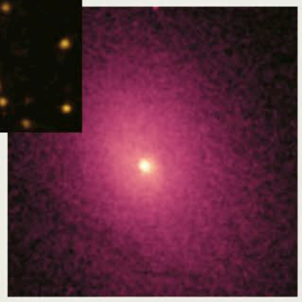


ABELL 2029
This visible-light image of Abell 2029 shows that it is an old, regular, spherical cluster full of elliptical galaxies.

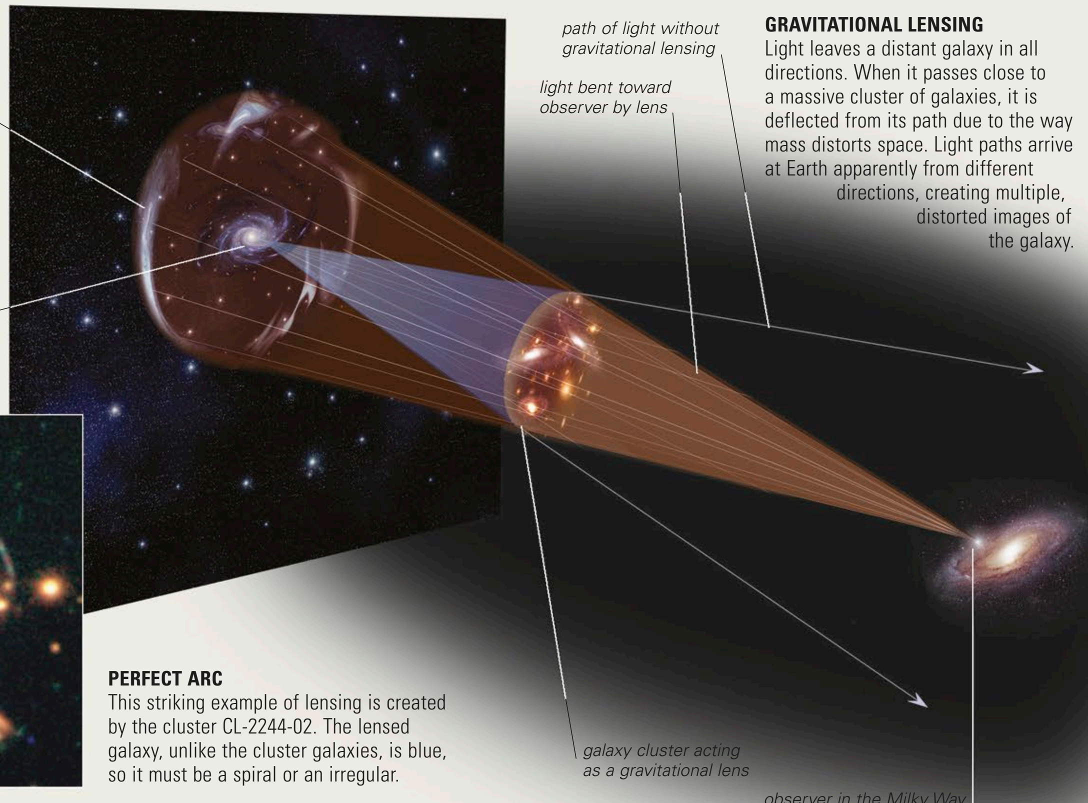


INTERGALACTIC GAS
An X-ray image of cluster Abell 2029 shows the hot gas cloud around its center. If not for the gravity of the cluster's dark matter, this gas would escape.

THE INTERGALACTIC MEDIUM

Astronomers can estimate the overall mass of a galaxy cluster from the way in which its galaxies are moving, but also through the phenomenon of gravitational lensing—an effect of general relativity (see pp.42–43). When a compact cluster lies in front of more distant galaxies, its mass bends the light passing close to it and deflects distorted images of the distant galaxies toward Earth. By measuring the strength of this effect, it is possible to measure the mass of the cluster and model how it is distributed. Galaxy clusters contain far more mass than the visible galaxies can account for, and most of it is in the matter that permeates the space between galaxies. This intergalactic medium is distributed around the cluster's center rather than around the galaxies.

X-ray satellites such as Chandra have revealed the nature of part of this material—large galaxy clusters often contain huge clouds of sparse, hot gas glowing at X-ray wavelengths. Most is hydrogen, but heavier elements are present. It is thought to originate in the cluster galaxies and to be stripped away during encounters and collisions. Most of a cluster's mass is not gas, however, but dark matter.



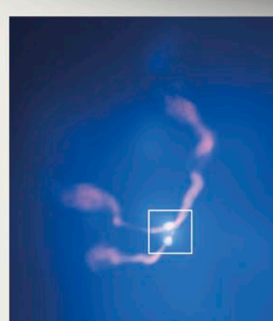
GRAVITATIONAL LENSING
Light leaves a distant galaxy in all directions. When it passes close to a massive cluster of galaxies, it is deflected from its path due to the way mass distorts space. Light paths arrive at Earth apparently from different directions, creating multiple, distorted images of the galaxy.

PERFECT ARC
This striking example of lensing is created by the cluster CL-2244-02. The lensed galaxy, unlike the cluster galaxies, is blue, so it must be a spiral or an irregular.

IRREGULAR AND RELAXED CLUSTERS
The central regions of the Virgo Cluster (above) and the Coma Cluster (below) show the difference between an irregular and a more spherical (relaxed) pattern of galaxies.

CLUSTER EVOLUTION

Astronomers have built a picture of cluster development that complements their models of galaxy evolution (see pp.306–309). According to their thinking, galaxy clusters start as loose collections of gas-rich spirals, irregulars, and small ellipticals. Because of their proximity and huge gravity, the spirals tend to merge, regenerating as spirals or forming ellipticals. Each interaction drives off more of the galaxies' free gas into the intergalactic medium. The high temperature and speed of atoms in this medium prevents their recapture by the cluster's galaxies. At this stage, the cluster is irregular, or "unrelaxed," and the pattern of galaxies and intergalactic gas is irregular and chaotic. However, as galaxies swing around each other, their random motions are eliminated, and they settle into a stable, spherical, "relaxed" distribution around the cluster's center. Eventually, even the largest elliptical galaxies begin to merge, forming giant ellipticals and cD galaxies. The hot gas, freed from ties to individual galaxies, sinks into the center of the cluster, where it lies evenly around the cluster's major elliptical galaxies. What remains is an old, spherical, relaxed cluster full of ellipticals.



VIOLENT MERGER
A composite image of Cluster Abell 400's core (left) shows two galaxies merging to form a giant elliptical. A composite image (below) shows surrounding X-ray (blue) and radio (red) emissions. Such a merger is typical of those that shape galaxy clusters.